

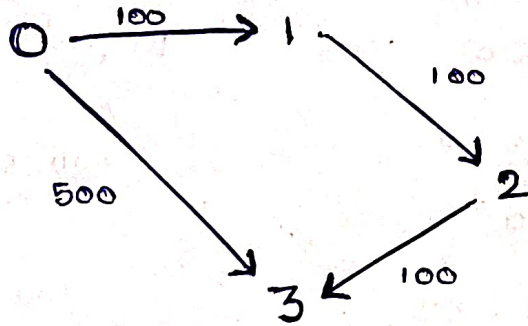
DRY RUN

Sample Input:

$n = 4$, flights = $[[0, 1, 100], [1, 2, 100], [2, 3, 100], [0, 3, 500]]$

src = 0 , dst = 3 , k = 1

Graph:



First initialize. Only source (0) is reachable initially.

dist = $[0, \infty, \infty, \infty]$

Iteration 1 [loop runs $k+1=2$ times]

temp = $[0, \infty, \infty, \infty]$

• Flight 1 (0 → 1, 100) dist[0] = 0 update
 $0 + 100 < \infty \rightarrow \text{yes}$ temp = $[0, 100, \infty, \infty]$

• Flight 2 (1 → 2, 100) dist[1] = ∞ No change
 cant' use

• Flight 3 (2 → 3, 100) dist[2] = ∞ No change
 cant' use

• Flight 4 (0 → 3, 500) $0 + 500 < \infty$ Update
 $\rightarrow \text{yes}$ temp = $[0, 100, \infty, 500]$

End of iteration 1

$\therefore \text{dist} = [0, 100, \infty, 500]$

Iteration 2

$$\text{temp} = [0, 100, \infty, 500]$$

• Flight 1 (0 $\rightarrow$ 1, 100)

$$0 + 100 < 100$$

No

No change

• Flight 2 (1 $\rightarrow$ 2, 100)

$$\text{dist}[1] = 100$$

$$100 + 100 < \infty \rightarrow \text{Yes}$$

Update

$$\text{temp} = [0, 100, 200, 500]$$

• Flight 3 (2 $\rightarrow$ 3, 100)

$$\text{dist}[2] = 200$$

$\therefore$ No

No change

• Flight 4 (0 $\rightarrow$ 3, 500)

$$0 + 500 < 500$$

No

No change

End of iteration 2

$$\therefore \text{dist} = [0, 100, 200, 500]$$

$\therefore$ The final answer $\Rightarrow \text{dist}[3] = 500$